CALM Lab useRguide

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2026-10-18

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Welcome

This is the useRguide for the CALM Lab. It is intended to be a resource for new lab members to get up to speed with the tools and workflows that we use in the lab. It is also a reference for existing lab members to refresh their knowledge or learn new skills.

What is R?

According to the R website

> R is a free software environment for statistical computing and graphics.

Let's unpack this.

- Free is nice!
- Statistical computing is a general term referring to a large class of techniques that involves statistics, machine learning, and more.
- Graphics refers to creating visualizations of data and output from statistical computing.

It's really all about **data**. We live in a world where data is everywhere. Our personal lives, the media, and our work are all filled with data. That data is hidden behind numbers stored in spreadsheets, databases, and other formats. A fundamental skill in the modern world is the ability to analyze and visualize data. R is a tool that can help you do just that.

R is a programming language. There are pros and cons to using a programming language over Excel and GUI (Graphical User Interface) applications with click-and-drag functionality. The pros are related to the fact that programming languages are more flexible, powerful, and reproducible than GUI applications. The cons are that programming languages have a steeper learning curve than GUI applications and there are often more than one way to do the same thing in a programming language (possibly leading to confusion).

Debates on which programming language you should learn and use are rampant on the internet and social media. Regardless of which programming language you learn, working with data through code provides a challenge that will open you up to the world of **data**. You will gain an appreciation for the level of mundanity and complexity that is involved in processing the vast amounts of data in this world. Through working with data yourself, you will also become a better consumer of data and information that you see in your own work, the news, and social media.

It all starts with gaining a basic level of proficiency in 1) working with data, 2) data visualization, and 3) statistical computation. The first few chapters of this guide will focus on these three areas. After that, you will be able to explore more advanced levels of data analysis and visualization.

Install and Setup

First things first, you need to install R and Positron and configure your environment in a way that is conducive to our workflow in the lab.

If you do not have R or Positron installed on your computer, then follow the install instructions below.

If you already have R or Positron installed on your computer, then you may want to update your installations.

Install Software

Install R

Download and install R from the R Project website.

1. Go to the R Project website.
2. Click on the "CRAN" link under "Download, Packages" in the left-hand menu.
3. Choose the 0-Cloud mirror option.
4. Select the **base** installation (for Windows) or the **Latest Release** (for Mac) option.
5. Open and run the installer file that you downloaded. Follow the prompts to complete the installation.

Install Positron

Download and install Positron from the Positron website.

1. Go to the Positron website.
2. Click on the "Download" button.
3. Select the appropriate version for your operating system.
4. Open and run the installer file that you downloaded. Follow the prompts to complete the installation.

Update Software

Update R

Warning: When updating R (not RStudio), it may remove all packages you have installed.

First check what version of R you have installed by opening Positron and checking the R version number in the console when you start it up. If you have an R version older than 4.6.0, then you should update your R installation.

Use the installr package to update R on Windows. To do this, run the following code in the Positron console:

```
install.packages("installr")
library(installr)
updateR()
```

Update Positron

To update Positron, go to the Positron website and download the latest version for your operating system. Then, open and run the installer file that you downloaded. Follow the prompts to complete the installation.

Getting to Know Positron

While R is the programming language that we will be using, Positron is the software that we will use to write and run R code. Positron is an Integrated Development Environment (IDE) for R. It provides a user-friendly interface for writing and running R code, as well as tools for debugging and visualizing data.

Open Positron and go through the Welcome Tour and Setup Guides to get familiar with the interface and features.

Configure Positron

Have a lab member or Dr. Tsukahara walk you through the Positron interface, and help you configure the settings to match the lab's workflow. The following sections provide a summary of the recommended settings.

Install Extensions

To install an extension, click the **Extensions** icon (left sidebar) and search for the extension name.

Feel free to explore the available extensions, but the following are recommended for our lab workflow:

- **Fira Code** — a fancy monospaced font with ligatures (e.g., `!=` becomes).
- **Material Icon Theme** — icon themes change the look of folders and files.
- **One Dark Pro** — a dark theme for your IDE.
- **GitHub Theme** — a light theme for your IDE.
- **vscode-pets** — adds a cute pet to your IDE.

Opening Settings

- Press Cmd, or click the gear icon (bottom-left) → **Settings**.
- In the Settings window, click the **Search settings** bar at the top and type in the name of the setting you want to change. The list of settings will filter as you type.

Files & whitespace

Setting	Search for	Action
Files: Associations	`files associations`	Click **Add Item** → Item: `renv.lock`, Value: `json`
Files: Trim Trailing Whitespace	`trim trailing`	Check
Files: Insert Final Newline	`insert final newline`	Check
Files: Trim Final Newlines	`trim final newlines`	Check

Editor

Setting	Search for	Action
Editor: Font Family	`font family`	Set to `Fira Code` (keep existing fallbacks after it)
Editor: Font Ligatures	`font ligatures`	Check
Editor: Font Size	`editor font size`	Set to `14`
Editor: Format On Type	`format on type`	Check
Editor: Tab Size	`tab size`	Set to `2`

Setting	Search for	Action
Editor: Detect Indentation	`detect indentation`	**Uncheck**
Editor: Auto Indent	`auto indent`	Dropdown → `full`
Editor: Wrapping Indent	`wrapping indent`	Dropdown → `indent`
Editor: Default Formatter	`default formatter`	Dropdown → **Positron R** (`positron.positron-r`)

Quarto

Setting	Search for	Action
Quarto › Cells › Background: Color	`quarto background color`	Dropdown → `useTheme`
Quarto › Visual Editor: Markdown Wrap	`quarto markdown wrap`	Dropdown → `sentence`
Quarto › Visual Editor: Default List Spacing	`quarto list spacing`	Dropdown → `tight`

Positron & keybindings

Setting	Search for	Action
Positron › Assistant: Enable	`positron assistant enable`	Check
Workbench › Keybindings: Rstudio Keybindings	`rstudio keybindings`	Check

Workbench, window & appearance

Setting	Search for	Action
Workbench: Startup Editor	`startup editor`	Dropdown → `readme`
Symbols: Hides Explorer Arrows	`hides explorer arrows`	**Uncheck** (needs Symbols extension)
Window: Restore Windows	`restore windows`	Dropdown → `none`
Window: Close When Empty	`close when empty`	Check

Icon theme (`vscode-icons`): set this via the picker rather than search — open the Command Palette (**Cmd/Ctrl+Shift+P**) → **Preferences: File Icon Theme** → choose **VSCode Icons**.

Git & GitHub

Setting	Search for	Action
Git: Autofetch	`git autofetch`	Check
Git: Enable Smart Commit	`smart commit`	Check
Git: Confirm Sync	`confirm sync`	**Uncheck**
GitHub › Copilot › Next Edit Suggestions: Enabled	`next edit suggestions`	Check (needs Copilot extension)

Special cases

Markdown-only whitespace override (`"[markdown]": { trimTrailingWhitespace: false }`): in the Settings search bar type `@lang:markdown`, find **Trim Trailing Whitespace**, and **uncheck** it. The funnel scopes that single change to Markdown only and writes the language block for you.

Editor: Rulers (`[80]`): this array has no graphical control. Searching for `rulers` only offers an **Edit in settings.json** link, so this one entry must stay in JSON — add `"editor.rulers": [80]`.

References